EdgeSat AI: compact land-cover classification under synthetic sensor noise

Track 3 — AI/ML in Space Technology | QAC-Net prototype

Summary. A small two-head CNN classifies 64×64 EuroSAT tiles and estimates whether an input is clean or synthetically corrupted. The model meets the size and measured Colab CPU-latency goals. It also reports an uncalibrated entropy-based ambiguity score. Clean macro-F1 and two of three noise robustness goals remain unmet.

1. Problem and formulation

Satellite downlink bandwidth is limited, so an onboard pipeline could rank image tiles for transmission, deferral, or review. This prototype addresses the land-cover classification and synthetic sensor-noise portion of the Track 3 challenge. For an input tile $x \in [0,1]^{3\times64\times64}$, the land-cover head estimates $p(y|x)$ over 10 EuroSAT classes; a second head estimates $q \in$ {clean, synthetic corruption}. A normalized softmax entropy is also reported for each head as an uncalibrated ambiguity indicator, not a calibrated probability of error. The quality head is not cloud detection. EuroSAT does not provide cloud labels.

2. Data and method

The organizer's shared subset procedure selected 2,000 RGB tiles from torchvision EuroSAT with seed 42. The official random split assigned 1,600 tiles to the train pool and 400 to the held-out test set; the train pool was divided into 1,440 fit and 160 validation examples (seed 43). The same test set was used for both runs. The compact QAC-Net uses depthwise-separable convolution blocks and two output heads. We trained one clean-only baseline and one noise-augmented variant. Evaluation adds Gaussian noise (σ=0.15), salt-and-pepper noise (severity 0.15), or drops one RGB channel.

Figure 1. Inference flow; the quality head is trained on clean vs synthetic corruption examples.

We report macro-F1 across the 10 land-cover classes, accuracy, per-class F1, the clean-to-corrupted macro-F1 drop, exported TorchScript size, and batch-1 CPU latency. Macro-F1 gives each class equal weight. F1 drops are absolute percentage-point (pp) differences.

3. Quantitative results

Held-out test set: 400 tiles. The handout states trade-offs, not thresholds; numeric targets below are team goals, not claimed organizer requirements.

Metric	Clean-only	Noise-trained	Project goal
Clean macro-F1	82.09%	77.05%	≥85%
Clean accuracy	83.25%	78.75%	—
Gaussian macro-F1 / drop	2.14% / 79.95 pp	59.75% / 17.30 pp	drop ≤8 pp
Salt-pepper macro-F1 / drop	1.69% / 80.39 pp	57.25% / 19.80 pp	drop ≤8 pp
Channel-drop macro-F1 / drop	9.01% / 73.07 pp	70.38% / 6.67 pp	drop ≤8 pp
TorchScript / CPU latency	0.449 MB / 2.23 ms	0.449 MB / 2.23 ms	<5 MB / ≤10 ms
Parameters	89,820	89,820	—

Interpretation. The clean-only model scores higher on clean macro-F1. Noise training improves scores under the three tested corruptions, but the Gaussian and salt-and-pepper drops still exceed the 8 pp goal. Channel-drop loss is within target. Both exports meet the <5 MB model-size criterion and measured Colab CPU latency is below 10 ms/tile. These latency results are specific to the reported Colab CPU runtime.

4. Takeaways and limitations

This is a reproducible baseline and demo, not a submission that meets every performance target. Neither model reaches 85% clean macro-F1. The quality head detects the synthetic clean/corrupt distinction it was trained on; it must not be presented as cloud screening. No real cloud-recognition result, uncertainty calibration/reliability analysis, flight-hardware measurement, downlink-volume saving, or end-to-end onboard policy has been demonstrated. A single local prediction (Pasture, 92.63% class confidence) illustrates inference only and does not establish aggregate accuracy.

Next: add the organizer’s CloudSEN12 mini-split with cloud labels; tune training and augmentation against a validation set; evaluate once on the unchanged held-out test set; then calibrate and evaluate uncertainty on validation data. Measure latency and power on intended edge hardware. Report per-class metrics and reliability for any revised model.

5. Reproducibility

The accompanying submission code includes the Colab notebook, fixed split manifest, training/evaluation scripts, configuration, checkpoints, TorchScript export, and results JSON. Re-run the notebook with the organizer’s seed-42 2,000-tile subset and preserve the 1,600/400 split for comparison. Dataset source: EuroSAT via torchvision.datasets.EuroSAT; subset and split protocol follow the participant handout, Track 3.